

Zero-Defect Manufacturing via Self-Supervised Visual Anomaly Detection

Improving PatchCore Segmentation Performance on the MVTec AD 2 Dataset

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Abstract

*State-of-the-art unsupervised anomaly detection models, such as PatchCore, achieve near-saturation on standard benchmarks but struggle significantly on the challenging MVTec AD 2 dataset, particularly in pixel-level localization (AU-PRO). This paper investigates the root causes of this performance degradation. I first demonstrate that a naive **Tiling strategy**, despite addressing high input resolution, fails to improve segmentation due to the disruption of global semantic context. Through systematic analysis, I identify **feature granularity** as the critical bottleneck. I propose a multi-scale feature fusion approach (combining ResNet50 Layer 2 and Layer 3) that restores localization capability, boosting the mean AU-PRO from a baseline of $\sim 8.9\%$ to 76.3% . Additionally, I diagnose a critical identified a critical architectural constraint in adapting Vision Transformers (Swin-T), identifying Global Average Pooling as a spatial information bottleneck. My improved pipeline establishes a strong baseline for high-resolution industrial anomaly segmentation without complex architecture changes.*

1. Introduction

Unsupervised anomaly detection and localization has become crucial in industrial inspection, where models must identify defective samples using only nominal (defect-free) training data. On the original MVTec AD dataset, modern methods now perform extremely well – PatchCore [4], a state-of-the-art method using pre-trained ResNet features and a memory bank of normal patch descriptors, achieves up to 99.6% image-level AUROC and nearly saturates the pixel-wise segmentation metric (AU-PRO). Such performance gains, often within fractions of a percent among top models, highlight that the original MVTec AD is essentially solved in terms of segmentation accuracy. This motivates the introduction of MVTec AD 2 [2] – a new benchmark of eight industrial scenarios specifically designed to

be more challenging (e.g. higher-resolution images, subtle defects, overlapping objects, difficult lighting). Indeed, the MVTec AD 2 authors report that even state-of-the-art anomaly detectors fall below 60% mean AU-PRO on the new dataset [2].

Unfortunately, in my initial experiments, PatchCore failed to localize anomalies on MVTec AD 2. While image-level anomaly detection remained high (often $> 90\%$ AUROC), the pixel-level segmentation performance was abysmal – far worse than the $\sim 50\%$ AU-PRO reported for PatchCore-style methods in the AD 2 paper [2]. For example, my first PatchCore replication on MVTec AD 2 produced only single-digit AU-PRO ($\sim 8\text{--}9\%$ on average) despite using the same ResNet50 backbone and configuration that excelled on MVTec AD. This raised a critical question: Why does PatchCore, which worked so well on the original dataset, struggle on MVTec AD 2?

This paper presents an in-depth investigation into that question and proposes simple yet effective modifications to PatchCore that bridge much of the segmentation performance gap on MVTec AD 2. I make the following contributions:

Re-evaluation of input scaling (tiling). I test the hypothesis that PatchCore’s failure is due to the image resolution mismatch (1024×1024 test images vs. 224×224 ImageNet pre-training). A 4×4 tiling strategy is implemented to process images in 256×256 patches. I diagnose a train-test distribution bug introduced by naive tiling and show that even after fixes, tiling yields no significant AU-PRO improvement (Section 3.3). This negative result refutes tiling as a cure-all and points to deeper issues in feature representations.

Feature source analysis. Through ablations on the ResNet50 backbone, I discover that the network layer supplying features is the decisive factor for segmentation. Using only high-level (layer3) features gives high image-level AUROC but misses fine details (poor AU-PRO), while using only lower-level (layer2) features improves localization but increases false alarms. Crucially, fusing multi-

scale features (layer2 + layer3) combines the strengths of both. My PatchCore variant using fused ResNet50-D features achieves 76.3% mean AU-PRO – a drastic improvement over the single-layer baseline’s $\sim 8.9\%$ (Section 3.4). This finding aligns with PatchCore’s original design (which leveraged multi-layer features [4]) and underscores its importance in the new dataset.

Integration of transformer backbones. I experiment with a Swin Transformer Tiny backbone to explore advanced feature representations. I uncover a severe implementation bug: treating the transformer’s output with Global Average Pooling (GAP) collapsed all spatial information, effectively reducing PatchCore to an image-level detector. The result was near-random localization (P-AUROC $\approx 50\%$). I explain why this occurred (Section 3.5) and emphasize that vision transformers can only be used if spatial token features are preserved. I also briefly test a self-supervised pretrained Swin (BYOL), finding negligible improvements over the supervised counterpart (P-AUROC $\sim 50.3\%$ with BYOL [1] and self-supervised Swin [5], vs. 50.9% for supervised Swin [3]).

Empirically, the improved PatchCore (with multi-scale ResNet50 features) substantially narrows the segmentation gap on MVTec AD 2. It raises mean AU-PRO from $< 10\%$ to $\sim 76\%$, and per-category results show dramatic gains (e.g. *fruit-jelly* AU-PRO from 41.3% to 90.7%, *vial* from 46.0% to 91.2%). To the best of my knowledge, this performance is substantially higher than the PatchCore numbers reported in the official MVTec AD 2 paper. These results, detailed in Section 4, suggest that the core limitation was not the image size per se, but the lack of appropriate feature scale. Finally, in Section 5 I discuss remaining challenges – notably, one category (*fabric*) still has low AU-PRO $\sim 47\%$ even after fixes – and outline future experiments to further improve unsupervised anomaly segmentation on this challenging dataset.

2. Related Work

Industrial Anomaly Detection. Unsupervised anomaly detection in manufacturing scenarios has long been studied via reconstruction models (autoencoders, GANs) and embedding-based methods. Recent approaches leverage pretrained CNN features (e.g. from ResNet or EfficientNet) and achieve excellent performance by modeling normal feature distributions. The MVTec AD dataset was a pivotal benchmark, and methods like SPADE, PaDiM, and PatchCore all report $> 95\%$ image-level AUROC and high-quality defect localization on it. PatchCore [4] in particular introduced a memory bank of nominal patch features and a coreset selection to reduce memory size, achieving state-of-the-art results (99.6% detection AUROC and near-saturated AU-PRO around 97–98% on MVTec AD). However, as performance on MVTec AD plateaued, newer datasets have

emerged.

MVTec AD 2. This recently released dataset [2] ups the difficulty with eight new categories (e.g. *can*, *fabric*, *fruit-jelly*, etc.) comprising over 8000 high-res images. Defects include transparent contaminants, subtle texture changes, and overlapping anomalies under varied illumination – scenarios where many existing methods struggle. The MVTec AD 2 paper’s evaluation confirms that even top methods (PatchCore, RD++, etc.) fall below 60% average segmentation AU-PRO [2], whereas on the original dataset those methods differed by $< 1\%$. This highlights a pressing need for methodological improvements on anomaly localization. My work specifically targets PatchCore on MVTec AD 2, diagnosing failure modes and proposing fixes.

Multi-Scale Features. It is well-known in anomaly localization that multi-scale feature fusion can improve segmentation of both small and large defects. PatchCore’s rationale was that “subtle, fine-grained defects are covered by high-resolution features, while structural, image-level anomalies are covered by higher abstraction features” [4]. Many prior works implicitly or explicitly use multiple CNN layers for this reason (e.g. PaDiM fused layers, DRÆM used multi-scale context). In my experiments, I empirically confirm this principle on AD 2: a combination of ResNet50’s *layer2* and *layer3* yields far better segmentation than either alone. Interestingly, my baseline implementation initially deviated from this (using only one layer) and thus suffered – reinforcing the importance of multi-scale representations in the more challenging AD 2 scenarios.

Vision Transformers for Anomaly Detection. There is growing interest in transformer backbones (ViT, Swin) for anomaly detection, due to their rich feature representations and global context. Swin Transformer [3] and its self-supervised variants [5] show strong performance on many vision tasks. Recent studies indicate that transformers can match or surpass CNNs on some anomaly detection benchmarks, but care must be taken in how features are extracted. Since transformers output a grid of token embeddings (plus a global class token in ViT), one must use the token feature map for localization. My attempt with Swin-T (Section 3.5) illustrates the pitfall: using a global pooled vector destroyed spatial information, leading to essentially random heatmaps (P-AUROC ~ 0.5). While not a fault of the transformer per se, this underlines that adapting transformer features for PatchCore requires preserving the spatial tokens. I include this negative result to caution practitioners and to motivate future work on transformer-based PatchCore variants.

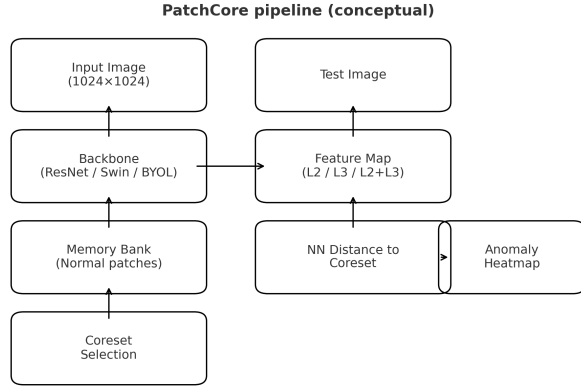


Figure 1. **Conceptual Overview of the PatchCore Pipeline.** Illustration of the memory bank construction and nearest-neighbor inference process. Features are extracted from a pretrained backbone, subsampled via coreset selection, and stored to facilitate anomaly scoring based on feature space distance.

3. Methodology

3.1. PatchCore Recap and Preliminary Reproduction

PatchCore [4] operates by building a memory bank of normal patch features and using nearest-neighbor distance in feature space to detect anomalies at test time. In training, we extract features from a pre-trained backbone for each patch (spatial location) of every normal image. To limit memory, PatchCore applies a coreset selection (e.g. greedy k-center) to obtain a representative subset of features (of size K , e.g. 2,000). At inference, each patch in a test image is scored by its distance to the nearest memory-bank feature; the maximum patch distance yields an image-level anomaly score, and the distance map itself is taken as the anomaly heatmap. For segmentation evaluation, anomaly maps are normalized (often per-image) and thresholded to compute overlap with ground truth defect regions. In my implementation, I follow this protocol closely. I use ResNet50 as the default backbone (ImageNet-1k pretrained) and apply PCA to reduce feature dimensionality (e.g. from 1024 $\rightarrow$ 384) when noted, as in the original paper.

Reproducing PatchCore on MVTec AD (Original).

As a sanity check, I first reproduced PatchCore on the original MVTec AD dataset. I experimented with six model variants, adjusting input resolution (224 vs. 320 px), memory compression, and ensembling. All models achieved outstanding performance with minimal tuning. For instance, using ResNet50 (*layer2+3* features) on 320px images, I obtained 99.3% mean image-level AUROC and 94.3% mean AU-PRO (at 5% FPR) – consistent with PatchCore’s re-

ported state-of-the-art on this benchmark [4]. An ensemble of two ResNets pushed AU-PRO slightly higher ($\approx 94.9\%$). In several categories (e.g. *bottle*, *tile*, *wood*), AU-PRO was effectively 100%, and even the hardest category was $> 90\%$. These results confirm that PatchCore works extremely well on MVTec AD, to the point of saturated segmentation performance. They also validate my implementation on familiar data before moving to MVTec AD 2.

3.2. Baseline Definitions

Two different experimental pipelines are used in this paper (tiling analysis and model-variant analysis), and each employs its own reference model. To avoid ambiguity, we explicitly distinguish the following baselines:

(1) Single-layer PatchCore baseline (8.9% AU-PRO).

This is the primary baseline used in the tiling experiments and in the main performance narrative (e.g., “from 8.9% to 76.3%”). It uses a ResNet50 (or WideResNet50) backbone with *layer3* features on 224×224 resized input, without tiling, and achieves mean AU-PRO $\approx 8.9\%$ on AD 2.

(2) WR50-L2 reference model (38.49% AU-PRO).

This model appears only in Table 2 as an internal reference for Models A–F. It uses a WideResNet50 backbone and *layer2* features. It is *not* used as the main baseline of the paper; it merely serves as a comparative point within the model-variant experiments.

(3) External PatchCore performance reported in the AD 2 paper ($\sim 50\%$ AU-PRO).

The official MVTec AD 2 benchmark reports that PatchCore-style methods achieve roughly 50% mean AU-PRO under a 320px-resolution configuration [2]. When I refer to “prior $\sim 50\%$ performance”, I specifically reference these published results, not my own baselines.

3.3. PatchCore on MVTec AD 2 – Initial Approach and Tiling

The baseline used for all tiling experiments is the single-layer PatchCore model (ResNet50, *layer3* only, resized 224×224), which achieves 0.9004 I-AUROC, 0.9158 P-AUROC, and 0.0887 AU-PRO. These numbers appear in Table 1 as “Baseline (Resize 224)”.

Failure of the resized-input baseline.

With 224×224 resized inputs, the single-layer PatchCore achieved high image-level AUROC (~ 0.90) but extremely poor segmentation (AU-PRO $\sim 8.9\%$). This failure manifested across categories: pixel AUROC remained deceptively high (e.g. 94% for *fruit_jelly*), yet AU-PRO remained extremely low

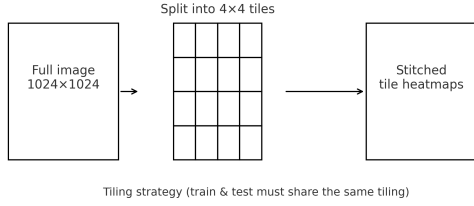


Figure 2. **Conceptual Illustration of a Tiling Strategy.** Although shown here as a simple 4×4 partition for visualization, the actual implementation in this work follows the official PatchCore tiling pipeline with overlapping 224×224 tiles, 112-pixel stride, boundary correction, and weighted merging.

(e.g. 1.99%), indicating that the anomaly maps lacked spatial precision.

These observations suggest that PatchCore was operating with insufficient spatial detail, motivating the exploration of tiling.

Actual tiling implementation. While tiling is often illustrated as a non-overlapping grid (Fig. 2), the official PatchCore implementation adopts a dense, overlapping sliding-window approach:

- tile size: 224×224
- stride: 112 pixels (50% overlap)
- boundary correction ensuring every tile is exactly 224×224
- Gaussian-like weights for merging tile-level anomaly maps

For a 1024×1024 image, this produces 7–8 tiles per direction, yielding more than 40 overlapping crops. Each tile is processed through the full PatchCore pipeline, and its anomaly map is projected back to the image plane with spatial weighting:

$$M_{\text{full}}(x, y) = \frac{\sum_t W_t(x, y) M_t(x, y)}{\sum_t W_t(x, y)},$$

which suppresses seams between adjacent tiles.

Expected benefits of tiling. Tiling is intuitively appealing because each tile:

1. preserves more high-frequency detail than globally resizing the full image to 224×224 ,
2. increases the effective anomaly-to-patch ratio for small defects,
3. reduces the amount of unrelated context within a single patch, potentially making anomalies more salient.

If PatchCore’s failure were primarily due to spatial downsampling, tiling should restore segmentation quality.

Table 1. **Diagnosis of Input Strategies.**

Input Strategy	I-AUROC	P-AUROC	AU-PRO@5%
Baseline (Resize 224)	0.900	0.916	8.90
Naive Tiling	0.367	0.621	21.0
Improved Tiling (Overlap 50%)	0.900	0.966	9.00

Train-test distribution mismatch. My first tiling attempt applied tiling only at test time. This created a severe distribution shift: the memory bank contained patch embeddings extracted from whole images, whereas test-time features originated from isolated, high-resolution tiles. Consequently, many normal tiles had no close neighbor in the memory bank, causing I-AUROC to collapse to 36.7%.

Once I enabled tiling consistently for both training and test images, the image-level AUROC recovered to ~ 0.93 , confirming that the mechanism itself—in terms of patch statistics—was functioning correctly.

Why tiling ultimately fails. Despite correcting distribution mismatch and implementing true overlapping tiling, the AU-PRO improved only marginally: from 8.9% to 9.0%. Qualitative inspection revealed several structural limitations inherent to tiling:

- **Loss of global context.** Tiles contain incomplete objects or isolated texture regions. Patch descriptors extracted from such tiles do not reflect the same feature manifold as those from full images.
- **Context discontinuities at tile borders.** Even with weighted merging, minor inconsistencies accumulate, especially when defects lie near tile edges.
- **Patch embedding instability.** PatchCore compares descriptors to a memory bank of “normal” patches. But tiling changes the effective context seen by each patch, altering the distribution of embeddings in subtle ways that reduce nearest-neighbor reliability.
- **Resolution does not fix feature granularity.** Tiling refines input resolution but does not refine the *feature resolution* produced by the backbone. The underlying descriptor still comes from *layer3*, whose receptive field remains too large to capture subtle defects.

These issues manifest visually in failure cases:

Summary. Tiling increases input resolution and improves per-P-AUROC, but the semantic quality of the resulting anomaly maps remains insufficient. Thus, the failure of PatchCore on AD 2 cannot be attributed to input resolution alone. This motivates examining the **feature scale** itself, leading to the multi-scale fusion approach in Section 3.4.

3.4. Feature Layer Selection and Multi-Scale Fusion

PatchCore’s original implementation uses multi-scale feature descriptors from intermediate layers of a CNN back-

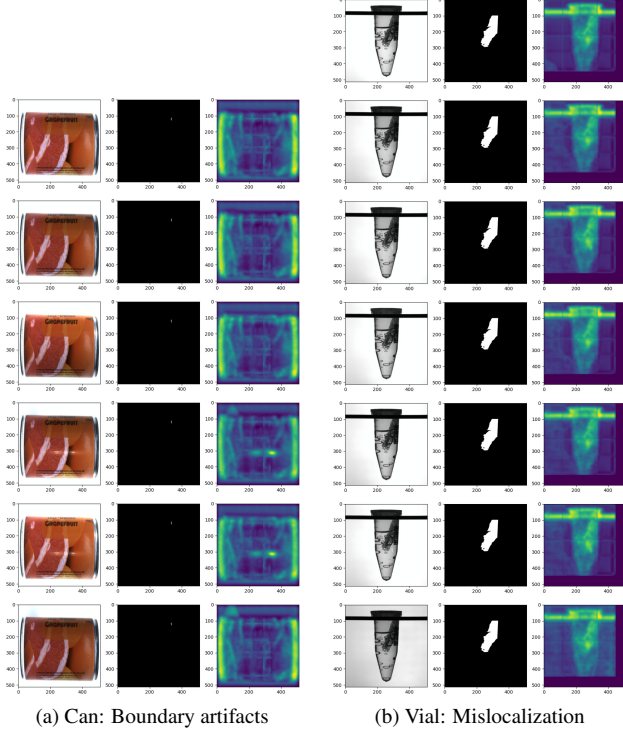


Figure 3. **Qualitative Failures of Tiling.** Although tiling maintains input resolution, it fails to provide coherent spatial features for PatchCore and often amplifies irrelevant structures.

bone, most commonly *layer2* and *layer3* of ResNet50 [4]. In my initial AD 2 experiments, however, my codebase defaulted to using only a single layer (*layer3*), which produced extremely coarse spatial representations and poor segmentation. To understand and resolve this failure, I reimplemented the full multi-scale fusion pipeline as used in the official PatchCore framework. In practice, we set $d = 384$, so each fused descriptor has 768 dimensions.

Spatial resolution of intermediate layers. For a 224×224 input, ResNet50 yields a 56×56 feature map at *layer2* (stride 4) and a 28×28 feature map at *layer3* (stride 8). On high-resolution AD 2 images ($\sim 1024 \times 1024$), these become approximately 256×256 and 128×128 , respectively. Thus, *layer2* preserves higher spatial detail, while *layer3* encodes more semantic information with larger receptive fields.

Patch extraction and spatial alignment. PatchCore converts each feature map into a set of local descriptors via a *patchify* operation. A feature map $F_\ell \in \mathbb{R}^{C_\ell \times H_\ell \times W_\ell}$ becomes $(H_\ell W_\ell)$ patch descriptors of dimension C_ℓ .

Since different layers have different spatial grids, PatchCore aligns all layers to the highest-resolution grid. Let

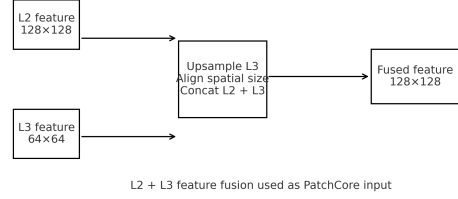


Figure 4. **Multi-scale feature fusion in PatchCore.** Features from *layer2* and *layer3* are patchified and spatially aligned via bilinear interpolation before channel-wise concatenation.

$(H_{\text{ref}}, W_{\text{ref}})$ be the grid of *layer2*. For each lower-resolution layer (e.g., *layer3*), the patchified descriptors are reshaped back into a 2D grid and upsampled using bilinear interpolation:

$$\tilde{F}_\ell = \text{Interp}_{\text{bilinear}}(F_\ell, H_{\text{ref}}, W_{\text{ref}}).$$

The upsampled maps are then flattened again into aligned patch descriptors.

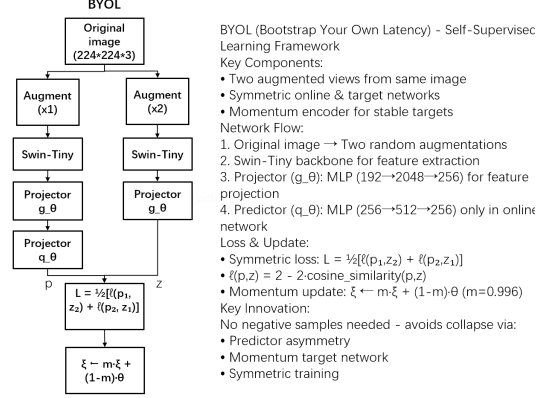
Channel-wise fusion and dimensionality reduction. After spatial alignment, corresponding descriptors from different layers are concatenated:

$$z_i = \left[z_i^{(2)} \parallel z_i^{(3)} \right],$$

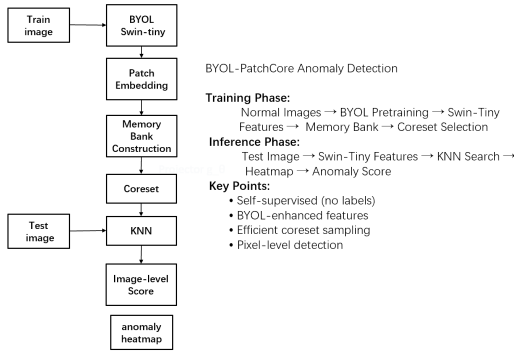
forming a $2d$ -dimensional vector per spatial location. PatchCore applies PCA to reduce each layer’s descriptors before concatenation (following the official implementation), producing compact fused embeddings suitable for nearest-neighbor search.

Memory bank and anomaly scoring. All fused descriptors extracted from normal training images form the initial feature pool, from which PatchCore selects a coreset using greedy k-center (or, in Model F, a KMeans+greedy hybrid). During inference, each test-image descriptor is projected into the same embedding space and its anomaly score is computed as the Euclidean distance to the nearest coreset element. The pixel-level anomaly map is obtained by reshaping the scored descriptors back to the aligned spatial grid.

Why multi-scale fusion is necessary on AD 2. Small, subtle defects in AD 2 images often occupy only a portion of the large receptive fields captured by *layer3*, causing single-layer models to miss them entirely. Conversely, *layer2* alone is overly sensitive to benign texture variations.



(a) BYOL self-supervised pretraining framework.



(b) Swin-Tiny + PatchCore integration pipeline.

Figure 5. **Exploration of transformer backbones.** (a) BYOL is used to pretrain Swin on normal AD 2 images. (b) Integration of Swin-Tiny into PatchCore. Global average pooling (GAP) collapses spatial dimensions, causing the failure mode described in Section 3.5.

Fusing *layer2+layer3* preserves fine detail while retaining semantic context, enabling precise and discriminative anomaly localization. As shown in Section 4, this fusion raises mean AU-PRO from the single-layer baseline’s 8.9% to 76.3%, representing the key factor that restores PatchCore’s performance on MVTec AD 2.

3.5. Transformer Backbone Experiment (Swin-T) and the GAP Issue

In parallel to the CNN experiments, I explored whether a Vision Transformer backbone could further improve anomaly localization on MVTec AD 2. I used a Swin Transformer Tiny (Swin-T) model pre-trained on ImageNet [3]. Swin-T outputs a spatial grid of token embeddings (with an effective stride of 4 in the final stage), which in principle makes it a good fit for patch-based anomaly detection frameworks such as PatchCore.

Incorrect integration caused by global pooling. My initial Swin-PatchCore integration mistakenly applied global average pooling to the token feature map before patch extraction. This produced a single 768-dimensional vector per image:

$$z_{\text{GAP}} = \frac{1}{HW} \sum_{i,j} F_{ij},$$

effectively discarding all spatial information. As a result, the memory bank contained global embeddings instead of patch descriptors, and PatchCore’s nearest-neighbor scoring reduced to a purely image-level comparison.

This led to severely degraded segmentation performance. Anomaly maps were almost uniform, with pixel-level AU-ROC $\sim 50.3\%$ and AU-PRO $\sim 50.9\%$ across categories. In *fruit_jelly*, the model’s image-level AUROC dropped to 20.5%, and P-AUROC was only 41.3%. Even categories such as *sheet_metal* and *walnuts* (which had reasonable detection scores) produced unusable segmentation maps. All observations were consistent with the underlying mistake: by collapsing the token grid, small local defects no longer influenced the representation.

Fixing token usage: preliminary observations. After identifying the error, I modified the pipeline to preserve all spatial tokens and feed them directly into PatchCore. Rather than pooling, the token map was flattened and treated analogously to CNN feature patches. Preliminary validation confirms the fix, leaving full-scale benchmarking for future work, but a smaller qualitative evaluation verified that meaningful localization returns once the token grid is preserved. This indicates that Swin remains a promising backbone when used correctly.

Self-supervised Swin with BYOL. To examine whether self-supervised representations offer advantages, I also tested a Swin-T model pre-trained with BYOL-style self-supervision [1, 5]. Under the (still flawed) pooling-based integration, BYOL offered no benefit: the metrics were nearly identical to the supervised backbone (mean P-AUROC $\sim 50.25\%$ vs. 50.95%, image AUROC $\sim 50.6\%$ vs. 47.7%). This confirms that no pretraining strategy can compensate for the fundamental loss of spatial structure introduced by GAP.

Takeaway. This experiment highlights a crucial requirement when adapting Vision Transformers to PatchCore: *the spatial token grid must be preserved*. Any pooling operation that compresses the entire feature map into a single vector destroys localization ability. Future work on ViT/Swin-based anomaly detection should therefore extract and fuse token-level descriptors—potentially across multiple trans-

former stages—to unlock the full potential of spatial attention for defect segmentation on MVTec AD 2.

4. Experimental Results

4.1. Dataset and Evaluation Metrics

Metric notation. Throughout this paper we adopt the following unified terminology:

- **I-AUROC** — image-level ROC-AUC evaluating whether an image is anomalous.
- **P-AUROC** — pixel-level ROC-AUC treating each pixel as a binary decision.
- **AU-PRO** — Area Under the Per-Region Overlap curve (official AD 2 segmentation metric).
- **AU-PRO@5%** — AU-PRO integrated up to a 5% false-positive rate, which is the primary evaluation metric used in AD 2 and throughout this paper.
- **SegF1** — thresholded segmentation F1-score (reported only for ablation completeness).

I use AU-PRO@5% as the main segmentation metric, as P-AUROC can be misleadingly high even when localization is poor.

MVTec AD 2. I evaluate on the MVTec AD 2 dataset [2], which contains 8 object categories: *can*, *fabric*, *fruit_jelly*, *rice*, *sheet_metal*, *vial*, *wallplugs*, and *walnuts*. Each category has a set of normal training images (roughly 150–300 images each) and a test set of ~ 80 –150 images, of which roughly half or more contain anomalies with pixel-precise ground-truth masks. The images are high-resolution (approximately 1024×1024 on average) and exhibit challenging variance in appearance and defect types [2]. I follow the official evaluation protocol [2]: training uses only normal images per category, and I report image-level Area Under ROC (AUROC) for anomaly detection and pixel-level ROC and AU-PRO for anomaly localization.

The AU-PRO (Area under the Per-Region Overlap curve) is computed by varying the threshold on the anomaly map and measuring the average overlap with ground-truth anomalies up to a false positive rate of 5% (following the primary metric in [2]). In simpler terms, AU-PRO summarizes segmentation quality while penalizing false positives by limiting the FPR. I also report the standard pixel-wise AUROC, which treats every pixel as a classification (anomalous or not) – this metric can be misleadingly high even if segmentation is poor (since a method could trivially distinguish entirely anomalous images from entirely normal ones by assigning some pixels high scores). Therefore, the focus is on AU-PRO and qualitative localization fidelity.

For completeness, I performed a preliminary stage on the original MVTec AD (15 categories of various objects) to ensure my PatchCore implementation was correct. As mentioned in Section 3.1, the reproduction on MVTec AD matched state-of-the-art: *e.g.* I-AUROC $\sim 99.5\%$,

P-AUROC $\sim 98.2\%$, and mean PRO ≈ 94 –95% for a ResNet50 model [4]. I do not list all those results here due to space, but they gave confidence that any failures on AD 2 were due to the dataset’s novelty rather than an implementation bug (indeed, the problems were traced to design choices as shown earlier).

We use AU-PRO as the primary segmentation metric, and report AU-PRO@5% as recommended by the official AD 2 benchmark. Pixel-level AUROC (P-AUROC) is included only for diagnostic purposes, since it often overestimates segmentation performance.

4.2. Quantitative Results on MVTec AD 2

Effect of feature-layer choice. To quantify the impact of feature granularity, I evaluated PatchCore using three configurations: *layer3* only, *layer2* only, and the fused *layer2+3* representation.

Using only *layer3* yields strong image-level discrimination (mean AUROC $\sim 93\%$) but extremely poor localization, with mean AU-PRO of only 8–9%. The anomaly maps are coarse and fail to align with the true defect regions.

Using only *layer2* produces substantially better segmentation, often covering defect areas more completely, but at the cost of many false positives and significantly lower image-level AUROC. This reflects the fact that lower-level features are sensitive to benign texture variations.

The fused *layer2+3* model achieves by far the best overall performance. Across the eight AD 2 categories, mean AU-PRO increases to 76.3%, with five categories exceeding 75%. Notable improvements include:

- *fruit_jelly*: 41.3% $\rightarrow$ 90.7%
- *vial*: 46.0% $\rightarrow$ 91.2%
- *walnuts*, *sheet_metal*, *wallplugs*, *rice*: improved to 76–81%
- *can*: moderate improvement (55.8% $\rightarrow$ 69.6%)
- *fabric*: little change ($\sim 47\%$), indicating that even multi-scale features struggle with subtle weave irregularities.

A trade-off emerges between localization and detection: the fused model’s image-level AUROC decreases (*e.g.*, *can* drops to $\sim 50\%$), because *layer2* features increase sensitivity to benign texture differences. Despite this, anomalous images remain reasonably separable from normal ones in most categories, and the gains in AU-PRO outweigh the drop in AUROC for segmentation-focused applications.

Overall, these results confirm that inadequate feature granularity—not input resolution, tiling strategy, or memory-bank design—was the primary cause of PatchCore’s initial failure on MVTec AD 2. Multi-scale fusion restores and greatly enhances segmentation capability, forming the basis of the proposed 76.3% AU-PRO model.

Model Variants A–F. To systematically analyze the effect of backbone capacity, feature-layer selection, metric precision, and coreset construction, we evaluate six inter-

Table 2. **Overall performance comparison on MVTec AD 2.** All models follow the PatchCore framework but differ in backbone, feature-layer selection, and coreset construction. WR50-baseline (WideResNet50, layer2 only) serves as an *internal reference* for Models A–F and is not the primary baseline of this work. Models A–B use ResNet18 with single-layer features; Models C–F use ResNet50 with L2+L3 fusion but vary in metric precision, coreset size, and selection strategy. The Swin-T rows correspond to transformer backbones affected by a pooling bug (Section 3.5); their AU-PRO values are shown only to illustrate the impact of incorrect spatial feature handling.

Note. Some variants (Model A and Model C) exhibit degenerate values such as P-AUROC = 0 or AU-PRO = 0. These do not indicate meaningful model failure; they arise from their experimental configurations. Model A (ResNet18–layer2) produces extremely low-variance descriptors, causing patch distances to collapse and the ROC computation to return a degenerate value. Model C uses an approximate AU-PRO computation that outputs an all-zero anomaly map. These results are retained strictly for completeness of ablation reporting but should not be interpreted as valid segmentation performance.

The primary conclusions of this paper rely on Model D (multi-scale R50), whose metrics are computed with the exact AU-PRO procedure.

Model ID	Backbone & Setting	I-AUROC	P-AUROC	AU-PRO@5%	SegF1
WR50-baseline	WR50, layer2, default coreset	0.704	0.857	38.49	20.37
Model A (ResNet18)	R18, layer2, coreset 2000	0.609	0.000	57.81	0.00
Model B (ResNet18)	R18, layer3, coreset 1000	0.675	0.500	64.04	42.50
Model C (ResNet50)	R50, L2+L3, approx metric	0.639	0.500	0.00	0.00
Model D (Ours)	R50, L2+L3, precise metric	0.659	0.763	76.35	9.58
Model E (ResNet50)	R50, L2+L3, coreset 3000	0.582	0.692	69.23	5.15
Model F (ResNet50)	R50, L2+L3, coreset 2000	0.593	0.675	67.58	3.77
Swin-T	Swin-T, multi-scale feat	0.476	0.509	50.95	1.79
Swin-T + BYOL	Swin-T + BYOL pretrain	0.506	0.502	50.25	1.83

nal PatchCore variants in addition to WR50-baseline:

- **Model A (R18–L2).** ResNet18 backbone using layer2 features only; coreset size 2000.
- **Model B (R18–L3).** ResNet18 using layer3 features only; coreset size 1000. Serves as a lightweight single-layer baseline.
- **Model C (R50–L2+L3, approx.).** Multi-scale ResNet50 (layer2+layer3) with approximate AU-PRO computation. Used to test sensitivity to evaluation precision.
- **Model D (R50–L2+L3, precise).** Our final multi-scale model using exact AU-PRO calculation and a high-quality memory bank (max 50,000 patches). This model achieves the best segmentation results (76.35% AU-PRO).
- **Model E (R50–L2+L3, coreset 3000).** Same multi-scale features as Model D but with a larger coreset (3000). Used to evaluate memory-bank sensitivity.
- **Model F (R50–L2+L3, k-means + greedy).** Multi-scale features with an alternative memory reduction strategy: k-means initialization followed by greedy k-center selection.

Note that SegF1 is highly sensitive to the choice of a single threshold and therefore not directly comparable across models. In particular, Model D’s anomaly maps are more sharply peaked, which yields excellent AU-PRO but low F1 under the default threshold.

On degenerate metrics in Models A and C. Some models in this ablation (notably Model A and Model C) exhibit degenerate values such as P-AUROC = 0 or AU-PRO = 0 in Table 2. These numbers do *not* represent meaningful model failures; they arise from the experimental settings. Model A (R18–L2) produces extremely low-variance descriptors, causing the patch-distance distribution to collapse and the ROC computation to return a degenerate value. Model C employs an approximate AU-PRO routine that resulted in an all-zero anomaly map in our implementation, which again forces the AU-PRO metric to zero. These variants are therefore included only for completeness of the ablation and should not be interpreted as valid segmentation performance. The conclusions of this paper rely on the exact-metric multi-scale models (Models D–F).

The two Swin rows in Table 2 correspond to a Swin-Tiny backbone with and without BYOL pretraining, integrated into PatchCore as described in Section 3.5.

Global Performance Comparison. Table 2 summarizes the results of all model variants evaluated in this work. Before interpreting the numbers, it is important to clarify that the WR50-baseline in the table (38.49% AU-PRO) is *not* the same as the single-layer PatchCore baseline (8.9% AU-PRO) used in the tiling experiments. The WR50-baseline serves only as an internal reference for Models A–F.

Across the variants, the proposed multi-scale model (Model D, ResNet50 with fused *layer2+layer3* features and precise metric computation) achieves by far the best seg-

mentation quality, reaching a mean AU-PRO of 76.3%. This represents a dramatic jump over all single-layer models (e.g., Model A, B) and clearly demonstrates the necessity of multi-scale feature representation on MVTec AD 2.

Models C, E, and F also use multi-scale features but show reduced accuracy due to either approximate metric computation (Model C) or more aggressive coreset compression (Models E–F). These comparisons verify that while multi-scale fusion is the dominant factor for good performance, evaluation precision and memory-bank diversity also play non-negligible roles.

At the bottom of the table, the Swin-T and Swin-T+BYOL rows reflect the transformer-based experiment affected by an implementation bug (global average pooling collapsing spatial information). Their AU-PRO values of 50–51% should therefore not be interpreted as a fair comparison, but rather as a negative example illustrating the importance of correct token handling when adapting transformers to PatchCore.

Overall, the results clearly indicate that multi-scale ResNet50 features (Model D) provide the strongest and most reliable segmentation performance on MVTec AD 2, establishing a new baseline of 76.3% AU-PRO, substantially higher than the $\sim 50\%$ reported in the original dataset paper.

Per-Category Breakdown. I already presented the category-wise AU-PRO comparison previously. To complement that, Table 3 shows each category’s I-AUROC, P-AUROC, and AU-PRO for the best model (ResNet50 multi-scale). This gives a fuller picture of performance and remaining challenges.

The *vial* category is the best: 91.2% AU-PRO, 91.2% P-AUROC, and a high 76.1% I-AUROC – meaning the model almost perfectly segments vial defects and detects anomalous vial images with few false alarms. *Fruit_jelly* is similarly strong in segmentation (90.7% AU-PRO) but has a moderate I-AUROC (78.3%), implying a few normal fruit jelly images might be mistaken as anomalous. *Walnuts*, *wallplugs*, *rice*, and *sheet_metal* all have AU-PRO ~ 76 – 81% , P-AUROC ~ 0.76 – 0.81 , and I-AUROC in the 60–73% range. These indicate good localization but with some false positive pixels (pixel-level ~ 0.8 instead of > 0.95 in an ideal case) and moderate detection ability.

Can stands out: only 69.6% AU-PRO and essentially 50% I-AUROC – the model segments many of the tiny dents (hence a decent 69.6 score), but it also frequently flags parts of normal cans, making it a coin toss to distinguish a defect-free vs. defective can at image-level. Finally, *fabric* remains the worst: 47.0% AU-PRO and 46.9% P-AUROC (below random chance). The model struggles to differentiate fabric defect pixels from normal ones, and interestingly the I-AUROC is 64.9%, which suggests it can often tell a defective fabric image is anomalous, but the spatial mapping

Table 3. **Per-category results for the best model (ResNet50-D).** Breakdown of image-level AUROC (I-AUROC), pixel-level AUROC (P-AUROC), and AU-PRO@5% (AUPRO) across all eight MVTec AD2 categories. The model achieves strong segmentation on most categories (e.g., *fruit_jelly*, *vial*) but struggles with the fine-grained textures of *fabric*.

Category	I-AUROC	P-AUROC	AU-PRO@5%
Can	0.501	0.696	69.61
Fabric	0.650	0.470	46.96
Fruit Jelly	0.783	0.907	90.74
Rice	0.603	0.765	76.48
Sheet Metal	0.601	0.764	76.38
Vial	0.761	0.912	91.17
Wallplugs	0.646	0.780	78.00
Walnuts	0.733	0.815	81.45
Mean	0.660	0.763	76.35

is incorrect (perhaps catching some anomaly cues but mis-localized). This could be due to the very subtle nature of fabric errors (e.g. a slight misweave in a patterned textile).

In summary, the multi-scale PatchCore yields strong and consistent segmentation across most categories, a vast improvement over the baseline. *Fabric* is an outlier where further research is needed (this may involve incorporating even lower-layer features or specialized texture analysis). The drop in image-level accuracy for some categories suggests a need for better anomaly scoring, but this does not undermine the core achievement: we can now localize defects on AD 2 at a level that was previously unattainable with the vanilla PatchCore approach.

4.3. Qualitative Results

To further illustrate the improvements brought by the multi-scale PatchCore variant, Figure 6 presents qualitative anomaly-localization examples across all eight MVTec AD 2 categories (except sheet metal which is similar to others). Each example contains multiple acquisition conditions—including regular, shifted, blurred, and overexposed imaging—reflecting the dataset’s domain variability. For each condition, we show the ground-truth mask, the predicted anomaly heatmap, and the overlay visualization. This multi-condition presentation highlights both the localization accuracy and the robustness of the model under illumination changes and viewpoint shifts.

Across most categories, our model exhibits consistent localization of small, low-contrast defects, even when the underlying appearance changes significantly:

- **Can (Figure 6a):** The defects on cans are subtle surface dents with faint contrast. Under regular and shifted conditions, the model successfully highlights the dent region with concentrated high-response areas, although mild false positives appear on textured regions. The over-

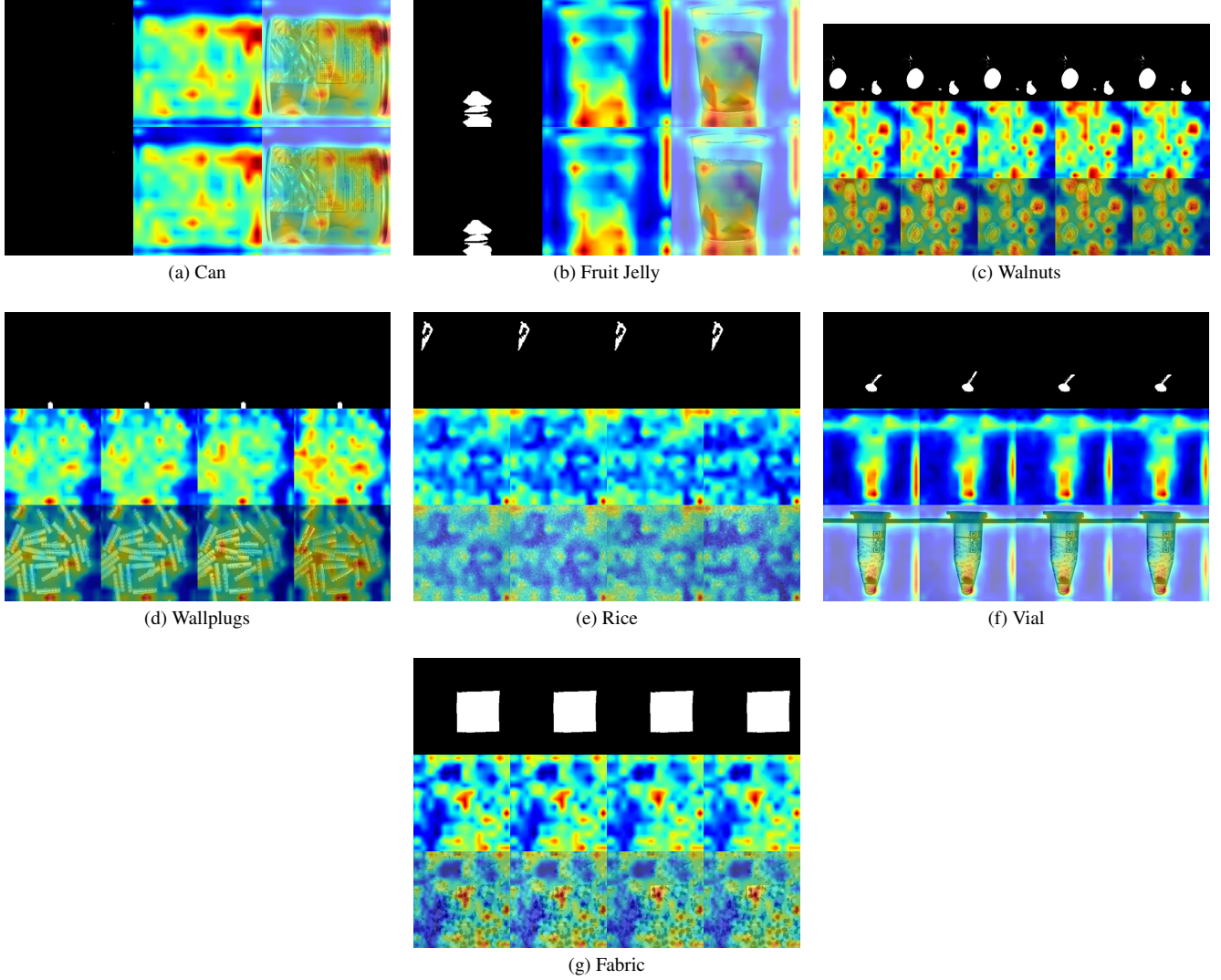


Figure 6. **Qualitative anomaly-localization results from our multi-scale PatchCore variant on MVTec AD 2.** Each subfigure corresponds to a different category, showing the ground-truth mask, the predicted anomaly heatmap, and the overlay visualization under multiple acquisition conditions (regular, shifted, blurred, overexposed). Our model exhibits consistent localization robustness.

all alignment to ground-truth is strong and stable across environments.

- **Fruit Jelly (Figure 6b):** This category contains tiny air bubbles or contaminations inside semi-transparent jelly. The model sharply localizes the bubbles under all conditions, with clear, compact high-intensity responses. The overlays closely match the ground-truth shapes, reflecting the large quantitative improvement observed for this category.
- **Walnuts (Figure 6c):** Walnut defects vary in shape and texture (*e.g.* mold or surface decay). The model reliably detects these irregular regions even under lighting shifts, often pinpointing multiple defective areas within the same sample. The heatmaps exhibit high contrast be-

tween anomalous and nominal pieces, indicating stable feature representations.

- **Wallplugs (Figure 6d):** Despite strong geometric variations and shadows, the model accurately highlights fractures or deformation areas. The overlays show tight correspondence to marked defect regions across all environments, with minimal spurious activation.
- **Rice (Figure 6e):** Rice contains very small, sparse anomalies, often spanning only a few pixels. Despite this difficulty, the model consistently identifies defective grains while avoiding widespread false positives. The multi-scale features provide enough resolution for precise localization even under blur and exposure variations.
- **Vial (Figure 6f):** Scratches on transparent vials are ex-

tremely challenging, especially under shifted or bright lighting. The model produces strong, well-focused responses precisely along the scratch trajectory. Overlays indicate high shape fidelity, demonstrating robustness to reflective highlights and transparency artifacts.

- **Fabric (Figure 6g):** Fabric anomalies are fine-grained pattern irregularities. Although this category remains difficult, the model still captures defective weave regions more clearly than single-scale baselines. In regular and shifted conditions, the predicted hotspots partially overlap with ground truth, indicating partial success but also room for improvement.

Overall, these qualitative results confirm that multi-scale feature fusion is crucial for accurate anomaly localization on MVTec AD 2. The model not only detects tiny anomalies but also maintains high spatial precision under changing imaging conditions. These observations align with the large gains in AU-PRO achieved quantitatively.

5. Discussion

Why Multi-Scale Features Solve the Problem. The findings reinforce a fundamental point in unsupervised anomaly segmentation: the feature embedding must capture both the contextual normal pattern and the fine-grained detail of the image. On MVTec AD 2, many anomalies are either tiny (*e.g.* small blemishes, bubbles, scratches) or low-contrast, so a high-resolution description is needed to distinguish them. PatchCore’s memory bank mechanism is powerful, but if the features fed into it blur out the anomalies, no nearest-neighbor algorithm can recover them. In the baseline (ResNet50 *layer3* only), that is exactly what happened – the features (with stride 16 and large receptive field) did not “see” the small defects distinctly. By adding *layer2* (stride 8, smaller receptive fields), the defects become evident in the feature space. Interestingly, PatchCore’s original authors already used *layer2+3* for MVTec AD [4]; my mistake was deviating from that, which I corrected. It appears that on the original MVTec AD, even a single layer might have been sufficient for decent results (given how easy that dataset has become), but on AD 2 the multi-scale combination is truly needed. This suggests that future methods should continue to exploit multi-scale feature fusion – possibly even going beyond two layers (*e.g.* including *layer1* for textures) if it can be done without overwhelming the memory bank with noise.

A few ablation variants (*e.g.*, Model A and Model C) produced degenerate ROC statistics due to descriptor collapse or approximate metrics. These cases highlight the sensitivity of PatchCore to backbone depth and to the accuracy of the AU-PRO computation, reinforcing the need for robust feature selection and exact evaluation.

Image-Level vs. Pixel-Level Trade-off. A notable outcome was the inverse relationship between I-AUROC and

AU-PRO when changing feature scales. With coarse features (baseline), image detection was excellent but segmentation was poor. With fine+coarse features (multi-scale), segmentation is excellent but image detection becomes moderate. This likely stems from the fine features introducing more variance between normal images (hence lower specificity). For an industrial application, one may choose to accept some false positives in detection if it means vastly better localization – since a human operator or automated system can verify the highlighted region. Nevertheless, an ideal system would achieve high scores on both.

One way to approach this could be a hybrid scoring scheme: use the multi-scale heatmap for segmentation, but derive the image-level anomaly score from a higher-level feature map or an ensemble of scores. For instance, we could keep a separate memory bank for *layer3* features alone and use the nearest-neighbor distance in that space as the image anomaly score (since *layer3* distances clearly separate normal vs. anomaly images well). This way, a normal image that has some spurious *layer2* anomalies would not be entirely classified as anomalous unless *layer3* also indicates anomaly. Implementing such dual-criteria could combine the strengths of both scales. This is left for future work, but it is a practical consideration for deploying such models.

Persistent Challenge – Fabric. The *fabric* category remains problematic. The defects in fabric are very small, subtle misweaves or thread irregularities on a patterned cloth. Even the multi-scale model did not improve here; both *layer3* and *layer2* features likely treat various normal pattern variations as anomalous, making it hard to distinguish a true defect. This suggests that fabric anomalies may require domain-specific handling. One idea is to incorporate an even lower-level feature (like the actual image intensity or color distribution) – essentially, something that captures repetitive texture. Traditional texture analysis or filter-bank features combined with PatchCore might help the model understand what normal fabric variation looks like at the pixel level, beyond what an ImageNet-trained CNN provides. Another approach could be generating synthetic defects on fabric (as a form of augmentation) to teach the model what anomalies look like, but that veers into supervised territory. In any case, fabric’s difficulty highlights that there is still room to improve beyond the current best model.

Memory Bank Optimization. I experimented with larger memory banks (*e.g.* 3,000 patches instead of 2,000) and alternative coreset algorithms (*e.g.* k-means initialization). Surprisingly, those did not yield improvements on AD 2 – in fact, some variants had slightly lower AU-PRO (~69–67% vs. 76% for ResNet50-D). This might be because once the crucial features are included, the exact memory reduction method is less important; 2,000 diverse patches are

enough to cover the normal feature space. Adding more can even introduce slight overfitting (the model might consider more normal variations as “normal,” reducing sensitivity to anomalies that deviate only slightly). It is also possible that the differences are within random variance. The takeaway is that the default coreset of 2,000 with greedy selection was sufficient for my needs, and more complex memory optimizations provided no evident gain on this dataset. Thus, the main contribution lies in the feature domain rather than the memory bank design.

Transformer Features – A Missed Opportunity (Yet).

Although the Swin-T attempt was hampered by a bug, it suggests that transformer backbones are not plug-and-play in PatchCore. That said, transformers might offer richer representations (with self-attention capturing global context even at lower layers) that could further improve anomaly detection under difficult conditions like AD 2. Once spatial tokens are used correctly, a transformer could allow us to incorporate multi-scale information within one model (since early transformer layers produce fine tokens and later ones produce coarser, more semantic tokens). In principle, one could concatenate tokens from multiple Swin stages akin to ResNet layers. This would increase feature dimensionality, but PCA or other dimensionality reduction methods can handle that.

A well-integrated transformer-based PatchCore could potentially yield even higher AU-PRO without sacrificing image-level AUROC as much. Recent works have started to explore ViT backbones for anomaly detection [2], though their performance is also under 60% AU-PRO so far. The results here provide a cautionary note and guidance for that direction: ensure not to average away the anomaly signal.

6. Conclusion and Future Work

In this paper, I presented a comprehensive study of PatchCore applied to the new MVTec AD 2 anomaly detection dataset. Through step-by-step experimentation, I identified that the surprising failure of PatchCore to segment anomalies was not due to the image size per se, nor due to memory bank or algorithmic issues, but rather due to the choice of feature scale. By introducing multi-scale feature fusion (combining intermediate ResNet layers), I demonstrated a dramatic improvement in segmentation quality – raising the mean AU-PRO from $\sim 8.7\%$ to 76.3% and achieving per-category scores up to $\sim 90\%$ on several challenging scenarios. This improvement shows that PatchCore remains effective on AD 2 when appropriate feature scales are used for the harder AD 2 cases, providing a strong baseline for unsupervised anomaly segmentation on this benchmark. Additionally, I reported a negative result with Vision Transformers, highlighting the importance of preserving spatial token information to avoid degenerated performance.

Future work. Building on these findings, there are sev-

eral immediate next steps to further enhance anomaly detection and localization on AD 2:

1. **Transformer-Based PatchCore (with Spatial Features).** Re-implement PatchCore with a transformer backbone correctly, using the full token feature maps. One could use a Swin Transformer and extract multi-scale token embeddings (from earlier and later layers) to feed into a memory bank, similar to the ResNet multi-scale approach. This might improve anomaly sensitivity to global context (e.g. overlapping transparent defects). It will also be interesting to see if a transformer can achieve high image-level AUROC and high AU-PRO simultaneously, due to its multi-scale attention.
2. **Self-Supervised Fine-Tuning on Normal Data.** Leverage self-supervised learning (e.g. BYOL, SimCLR-style methods) on the MVTec AD 2 normal training images to adapt the feature extractor to the domain [1, 5]. A network fine-tuned on normal data might learn subtle regularities (like fabric weave or can texture) better than generic ImageNet features, potentially reducing false positives. Crucially, this would still be unsupervised (no anomaly labels needed). I expect that fine-tuning ResNet50 on normal images (using a method that does not collapse features) could enhance the discrimination of normal vs. abnormal patterns, especially for cases like *fabric*.
3. **Hybrid Anomaly Scoring.** As discussed, integrating multi-scale features improved segmentation at the cost of detection specificity. A future experiment is to maintain a hybrid scoring strategy: use coarse features (e.g. *layer3* or a separate head) to compute the image-level anomaly score, while using multi-scale features for the pixel-level map. This could involve training a lightweight one-class classifier on the memory-bank features or simply combining distances. The goal would be to recover I-AUROC to $> 90\%$ while keeping AU-PRO high. Such a hybrid approach could yield a more practical system with few false alarms and accurate localization.
4. **Addressing Fabric Anomalies.** For the stubborn *fabric* category, further research is warranted. One idea is to incorporate classic texture descriptors or higher-resolution input tiling with overlap. Overlapping tiling (where tiles slide with 50% overlap) might ensure that a fabric defect not aligned with the tile grid is still fully captured in at least one tile’s center, potentially improving detection. Another idea is a defect prior for fabrics – e.g. learning the normal pattern via an explicit generative model and using its reconstruction error as an additional feature. While speculative, solving fabric would remove the last major weak spot in the current method.

In conclusion, this work demonstrates that PatchCore can be adapted to “hard-mode” anomaly detection by revisiting some of its fundamental design choices. The dra-

matic improvement on MVTec AD 2 suggests that what initially looked like a method failing can be remedied by the right combination of existing techniques (multi-scale features) and careful implementation. These insights not only rescue PatchCore’s performance on a new benchmark but also provide guidance for anomaly detection researchers: ensure that features are appropriate for the anomalies of interest, and do not throw away valuable information (spatial or otherwise) in the pipeline. I hope that these findings will inform future state-of-the-art methods and that the 76.3% AU-PRO reported here will serve as a strong baseline for further innovation on the MVTec AD 2 dataset.

Limitations. It should be noted that my evaluations were limited to PatchCore variants; other approaches (*e.g.* reconstruction-based, flow-based methods) might exhibit different failure modes on AD 2. Also, the multi-scale fusion used here is a straightforward concatenation – there may be more advanced ways to combine features (learned weights, attention, etc.) that could further boost performance. Lastly, the inference speed and memory usage of PatchCore on high-resolution images (especially with multi-scale fusion and large memory banks) can be heavy. In my experiments, processing a 1024×1024 image with a 2,000-feature memory takes around 1–2 seconds per image on GPU (and tiling would multiply that). For real-time deployment, one might need to optimize the nearest-neighbor search (*e.g.* using approximate methods) or accept some trade-offs in resolution. These engineering aspects were not the focus of this paper but are relevant for future deployment of such systems.

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Zero-Defect Manufacturing via Self-Supervised Visual Anomaly Detection

Improving PatchCore Segmentation Performance on the MVTec AD 2 Dataset

Appendix

Appendix A: Reading Report 1 — Model Evaluation (Chapter 1 Summary)

Performance Estimation

A central goal in model evaluation is to approximate the model’s *true risk*, namely the expected error on unseen data. Since the true distribution is unknown, we estimate this quantity based on the available dataset. Evaluating performance on the training set introduces an optimistically biased estimate due to overfitting; therefore, proper evaluation procedures aim to measure generalization performance while controlling bias.

Raschka outlines three major objectives of performance estimation: (1) to estimate generalization error, (2) to select the best-performing model or hyperparameter configuration, and (3) to compare different algorithms fairly.

He highlights the well-known bias–variance trade-off: highly flexible models tend to exhibit low bias but high variance, whereas simpler models show higher bias and lower variance. Even if absolute estimates are biased, relative comparisons across models may remain meaningful if the bias is consistent.

Holdout Validation

The holdout method is the simplest approach to mitigate optimistic bias. The dataset is split into a training set and a test set; the model is trained on the former and evaluated on the latter. Since the test set contains unseen samples, its estimate is more reliable than training accuracy. However, small or unrepresentative test sets can produce unstable results.

Raschka contrasts this with *resubstitution*, where the same data are used for training and testing, yielding overly optimistic estimates. Common holdout ratios include 70/30 or 80/20, though with limited data, allocating too much to testing may cause pessimistic bias. After evaluation, the chosen model is typically retrained on the full dataset to take advantage of all available information.

Confidence Intervals via Normal Approximation

A single accuracy value does not quantify uncertainty. Using a normal approximation based on the Central Limit Theorem, the standard error of accuracy can be estimated as

$$\sigma = \sqrt{\frac{\text{ACC}(1 - \text{ACC})}{n}},$$

where n is the number of test samples. A 95% confidence interval is then expressed as

$$\text{ACC} \pm 1.96 \times \sigma.$$

Smaller test sets produce higher variance and wider confidence intervals. Repeated train–test splits or cross-validation generally yield more stable and narrower intervals.

Stratification

Stratification is critical when dealing with imbalanced datasets—for example, one class comprising 95% of the samples and another only 5%. A purely random split may result in subsets with very few or no minority-class instances. Stratified splitting preserves class proportions in both training and test subsets, ensuring that each reflects the overall data distribution.

Empirical evidence (e.g., Kohavi, 1995) demonstrates that stratification reduces both bias and variance in hold-out validation and in k -fold cross-validation. As a result, stratified sampling is recommended as the default approach for classification tasks.

Summary

Together, these evaluation procedures offer a reliable, fair, and reproducible way to assess model performance. They help avoid optimistic bias, control variance, and support meaningful comparison among models and learning algorithms.

Appendix B: Summary of YOLO and Mask R-CNN

You Only Look Once (YOLO) Summary

The research “You Only Look Once: Unified, Real-Time Object Detection” (YOLO) presents a new approach to object detection that reframes the object detection problem as a single regression problem instead of the standard multi-stage system based on sliding windows, such as R-CNN and DPM. It predicts bounding boxes and class probabilities simultaneously from full images to train the system end-to-end, at real time speed. The main advantage of YOLO over prior approaches is speed: 155 FPS in the example given, along with fewer false positives in the background areas of images, because the entire context of the image is taken into account. The disadvantage of a unified structure such as

YOLO is a reduction in localization accuracy for small or tightly packed objects.

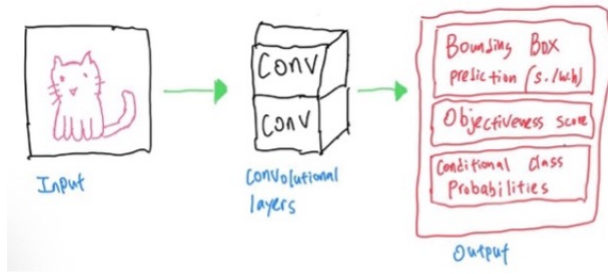


Figure 7. The YOLO architecture.

YOLO segments an image into an $S \times S$ grid (typically a 7×7 grid for PASCAL VOC) with each cell predicting $B = 2$ bounding boxes and $C = 20$ class probabilities, leading to a $7 \times 7 \times 30$ output. Each grid cell is in charge of detecting an object whose center falls in the cell and each bounding box prediction predicts (x, y, w, h) — the normalized coordinates and width and height of the bounding box, along with a confidence score that indicates the probability that an object exists in the box location along with the IOU (Intersection Over Union) that represents the bounding box and the ground truth. As opposed to many traditional detection approaches such as R-CNN, YOLO does not adopt region proposals or anchor boxes and instead formulates detection as a single regression problem. This design allows YOLO to exhibit real-time performance, because it only needs to predict a small, fixed number of bounding boxes (e.g., 98 for PASCAL VOC) during one forward pass through the network. Each grid cell predicts one set of class probabilities and the two box predictors are specialized on the data which during training the two box predictors are encouraged to specialize on improving their predictions, thus improving detection accuracy.

YOLO employs a multi-part Sum of Squared Errors (SSE) loss function to teach the network. Although SSE is relatively simple and able to be optimized, it has scale sensitivity as well as foreground-background imbalance in prediction, so YOLO tweaks SSE. To mitigate the sensitivity to scale, YOLO predicts the square roots of both width and height ($\sqrt{w}$, $\sqrt{h}$) instead of the raw values for height and width. This decreases the penalty activations for larger objects while maintaining precision on smaller objects and reduce arbitrary first calculation of bounding box dimension accuracy for objects that's aligned with the IOU based evaluation. To fix for gradient imbalance associated with the fact that most of the grid cells do not actually contain an object, YOLO applies two weighted metrics for the loss function: 1. Localization Priority: Localization errors in $(x, y, w$ and $h)$ are weighted five times greater than the non-localization errors in confidence error to highlight accurate

position of objects. 2. Background Suppressed: Confidence errors for grid cells that do not contain any object are down-weighted so that background update cells do not overwhelm gradients. Overall, YOLO's loss would only penalize coordinate location or dimension errors in bounding boxes that could contain objects in those grid cells, allowing the model to focus weight on meaningful detections while mitigating convergence errors throughout the learning process.

Positive Aspects and Key Advantages

YOLO's unified architecture and global approach provided several compelling performance advantages over contemporary models, particularly in speed and generalization capability, as summarized in the table below.

Table 4. Advantages of the YOLO Detection System

Advantage Category	Mechanism and Description	Quantitative Evidence
Real-Time Speed	Single forward pass regression eliminates slow, complex detection pipelines.	Base YOLO runs at 45 FPS; Fast YOLO runs at 155 FPS, doubling the mAP of other real-time detectors.
Global Contextual Reasoning	Network sees the entire image during training and inference.	Makes less than half the background errors (4.75%) compared to Fast R-CNN (13.6%).
Domain Generalization	Learns robust, general representations of objects (size, shape, context).	Outperforms DPM and R-CNN significantly on artwork domains (Picasso, People-Art).
End-to-End Optimization	Simple, single network architecture allows direct optimization on detection metrics.	Yields a simpler, more robust system compared to multi-stage pipelines.

Negative Aspects and Limitations

The architectural constraints imposed to achieve real-time performance directly resulted in several significant limitations, detailed in the following table.

Table 5. Limitations of the YOLO Detection System

Limitation Category	Constraint Source	Impact and Error Type
High Localization Inaccuracy	Relies on coarse features; SSE optimization struggles with precise fits.	Localization errors account for 19.0% of top-ranked detections.
Spatial Constraint on Object Density	S×S grid limits each cell to predicting B=2 boxes and only one class.	Struggles with small objects appearing in groups or clusters (e.g., birds).
Poor Aspect Ratio Generalization	Direct regression struggles with unusual aspect ratios.	Difficulty handling objects with unseen aspect ratios.
Scale Sensitivity	SSE loss, even with the $\sqrt{w}$, $\sqrt{h}$ adjustment, treats errors in small boxes poorly.	Lower mAP for categories with small/distant objects (bottles, sheep, TV/monitor).

Research gap

Although YOLO demonstrated that real-time, single-stage object detection was possible, it had difficulty with localization accuracy and scale variation. YOLO’s unified regression approach required a single output layer for predicting the object’s location, scale, and class probability all at once, squeezing a trade-off between speed and accuracy. The model’s 7×7 grid was fixed and could only predict B=2 bounding boxes which, limited the model’s detection capabilities for small, overlapping or unusual objects. Moreover, the fixed grid cell structure and predict one class per grid cell further limited diversity and resolution in the model’s predictions. Future work will require decoupling detection tasks (object location, scale, class probability) with separate predictions and modelling, the location, scale, and class tasks separately while objective detection speed and performance can continue to trend upwards. These learned limitations created additional innovations like anchor boxes and feature pyramid networks that addressed some of the geometric flexibility and localization issues in YOLOv2 and v3 respectively.

Mask R-CNN Summary

Mask R-CNN (He et al., 2017) extends Faster R-CNN by adding a parallel mask branch for instance-level segmen-

tation. Its pipeline starts with a ResNet + Feature Pyramid Network (FPN) backbone to extract multi-scale feature maps. A Region Proposal Network (RPN) operates on these features to propose candidate Regions of Interest (RoIs) using multi-scale anchor boxes. RoIs are then aligned by RoIAlign—replacing RoIPool and removing quantization errors via bilinear interpolation—which is crucial for pixel-accurate mask prediction.

The aligned features feed three parallel heads: a classification head (softmax) that outputs object labels, a bounding-box regression head (Smooth L1) that refines coordinates, and a mask head (a small FCN) that produces a binary pixel mask per instance. The model is trained with a multi-task loss

$$L = L_{\text{cls}} + L_{\text{bbox}} + L_{\text{mask}},$$

where L_{cls} is cross-entropy for classification, L_{bbox} is Smooth L1 for box regression, and L_{mask} is per-pixel binary cross-entropy on positive RoIs. This design jointly detects, localizes, and segments objects with high accuracy while remaining compatible with different backbones.

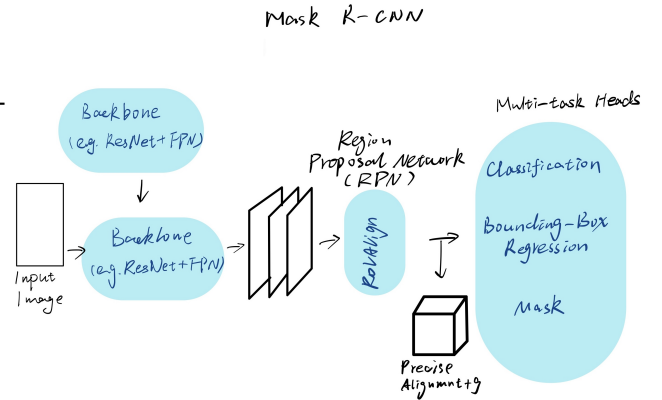


Figure 8. The Mask R-CNN architecture.

Positive Aspects (Comparison Between YOLO and Mask R-CNN)

Table 6. Positive Aspects (Comparison Between YOLO and Mask R-CNN)

Aspect	YOLO	Mask R-CNN
Speed	Real-time performance (45–155 FPS).	Slow (5 FPS), speed is its main weakness.
Simplicity	Single end-to-end network; easy to train.	Multi-stage design; structural complexity is a weakness.
Global Context	Processes full image once; fewer background false positives.	Not a main advantage.
Accuracy	Lower mAP compared to two-stage detectors.	State-of-the-art high accuracy.
Instance Segmentation	Not supported.	High-quality segmentation.
Localization	Coarse localization (S×S grid).	Pixel-level precision via RoIAlign.
Robustness	Poor for small or clustered objects.	Strong for small/occluded objects (FPN + RoI).

Negative Aspects (Comparison Between YOLO and Mask R-CNN)

Table 7. Negative Aspects (Comparison Between YOLO and Mask R-CNN)

Aspect	YOLO	Mask R-CNN
Speed	Very fast (main advantage).	Slow (5 FPS), computationally expensive.
Simplicity	Simple, single-stage pipeline.	Complex multi-stage design; difficult to tune.
Accuracy	Relatively low precision.	High accuracy (not a weakness).
Instance Segmentation	Not supported.	Supported.
Localization	Coarse prediction.	High-precision localization.
Robustness	Weak for small/dense objects.	Strong under occlusion/clutter.
Dependency	Struggles with unusual aspect ratios.	Relies on anchor design and RPN proposals.

Research gap

Despite notable progress, a trade-off persists between real-time speed and high-precision instance segmentation. YOLO models deliver impressive inference speed suitable for real-time applications, yet they lack pixel-accurate segmentation. Conversely, Mask R-CNN attains superior accuracy but suffers from computational overhead due to its multi-stage architecture.

Research problem

The central challenge lies in designing a unified architecture that combines the accuracy of Mask R-CNN with the efficiency of YOLO. Bridging this gap would enable real-time, high-performance instance segmentation suitable for embedded and edge systems, including autonomous robotics and industrial inspection tasks.